

HARDWARE FOR A2CMAN5

Given Data - $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$ $C = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$ C_{exp} Weights are preloaded in PE, ~~stay~~ and stay fixed,

TASK 1

$$PE = \begin{bmatrix} PE[0][0] = 5 & PE[0][1] = 6 \\ PE[1][0] = 7 & PE[1][1] = 8 \end{bmatrix}$$

TASK 2

C₀ (Initial-weights loaded, array empty).

$$\begin{matrix} 3 & 1 \\ 4 & 2 \end{matrix} \rightarrow \begin{bmatrix} PE_{00} & PE_{01} \\ PE_{10} & PE_{11} \end{bmatrix} = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

C₁ - Input 1 enters P[0][0].

$$\begin{matrix} 3 & & \\ 4 & 2 & \end{matrix} \begin{bmatrix} PE_{00} & PE_{01} \\ PE_{10} & PE_{11} \end{bmatrix} = \begin{bmatrix} 1 \times 5 = 5 & 6 \\ 7 & 8 \end{bmatrix}$$

C₂ - Psum flows down and 3 inputs.

$$\begin{matrix} 4 & 2 \\ & & \end{matrix} \begin{bmatrix} PE_{00} & PE_{01} \\ PE_{10} & PE_{11} \end{bmatrix} = \begin{bmatrix} 3 \times 5 = 15 & 1 \times 6 = 6 \\ 5 + (2 \times 7) = 19 & 8 \end{bmatrix}$$

C₃ - Psum flows down

$$\begin{bmatrix} PE_{00} & PE_{01} \\ PE_{10} & PE_{11} \end{bmatrix} = \begin{bmatrix} 5 & 3 \times 6 = 18 \\ 15 + 4 \times 7 = 43 & 6 + 2 \times 8 = 22 \end{bmatrix}$$

C₄ - psum 18 flows down.

$$\begin{bmatrix} PE_{00} & PE_{01} \\ PE_{10} & PE_{11} \end{bmatrix} = \begin{bmatrix} 5 & 6 \\ 7 & 18 + 4 \times 8 = 50 \end{bmatrix}$$

$$\underline{\text{Final}} = \begin{bmatrix} PE_{00} & PE_{01} \\ PE_{10} & PE_{11} \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

TABLE.

	R _{0in}	R _{1in}	P ₀₀	P ₀₁	P ₁₀	P ₁₁	Outputs
C ₀	-	-	-	-	-	-	-
C ₁	1	-	5	-	-	-	-
C ₂	3	2	15	6	19	-	C ₀₀ = 19
C ₃	-	4	-	18	43	22	C ₀₁ = 22 C ₁₀ = 43
C ₄	-	-	-	-	-	50	C ₁₁ = 50

TASK 3

- Total MAC operation = 8 MAC [2 MAC operations per output element (4 output element)]
- Input reuse - Each input is reused 2 ~~times~~ Times (Once per column).
- Off memory access - Total 12 Memory Access [4 reads for A, B and C each]

TASK 4

→ For output stationary array the partial sum of C would stay fixed in each PE. Each PE accumulates one output element locally while elements A and B stream through the array (typically A from left and B from top).